Chapter 12, Problem 28P

Problem

The line-to-line voltages in a Y-load have a magnitude of 880 V and are in the positive sequence at 60 Hz. If the loads are balanced with $Z_1 = Z_2 = Z_3 = 25\angle 30°$, find all line currents and phase voltages.

Step-by-step solution

Step 1 of 6

Refer to Figure 12.8 in the text book.

Calculate the value of the line current I_a.

$$I_a = \frac{V_{AB}\angle -30°}{\sqrt{3}Z_Y}$$

Substitute $880\angle 0°$ V for V_{AB} and $25\angle 30°\ \Omega$ for Z_Y.

$$\begin{aligned} I_a &= \frac{880\angle -30°}{\sqrt{3}(25\angle 30°)} \\ &= \frac{880\angle -30°}{43.30\angle 30°} \\ &= 20.32\angle -60° \text{ A} \end{aligned}$$

Therefore, the value of the line current I_a is $\boxed{20.32\angle -60° \text{ A}}$.

Step 2 of 6

Calculate the value of the current I_b.

$$I_b = I_a\angle -120°$$

Substitute $20.32\angle -60°$ A for I_a.

$$\begin{aligned} I_b &= (20.32\angle -60°)(1\angle -120°) \\ &= 20.32\angle -180° \text{ A} \end{aligned}$$

Therefore, the value of the line current I_b is $\boxed{20.32\angle -180° \text{ A}}$.

Step 3 of 6

Calculate the value of the current I_c.

$$I_c = I_a\angle 120°$$

Substitute $20.32\angle -60°$ A for I_a.

$$\begin{aligned} I_c &= (20.32\angle -60°)(1\angle 120°) \\ &= 20.32\angle 60° \text{ A} \end{aligned}$$

Therefore, the value of the line current I_c is $\boxed{20.32\angle 60° \text{ A}}$.

Step 4 of 6

Calculate the value of the phase voltage V_{an}.

$$V_{an} = \frac{V_{AB}\angle -30^\circ}{\sqrt{3}}$$

Substitute $880\angle 0^\circ$ V for V_{AB}.

$$\begin{aligned} V_{an} &= \frac{880\angle -30^\circ}{\sqrt{3}} \\ &= 508.07\angle -30^\circ \text{ V} \end{aligned}$$

Therefore, the value of the phase voltage V_{an} is $\boxed{508.07\angle -30^\circ \text{ V}}$.

Step 5 of 6

Calculate the value of the voltage V_{bn}.

$$V_{bn} = V_{an}\angle -120^\circ$$

Substitute $508.07\angle -30^\circ$ V for V_{an}.

$$\begin{aligned} V_{bn} &= (508.07\angle -30^\circ)(1\angle -120^\circ) \\ &= 508.07\angle -150^\circ \text{ V} \end{aligned}$$

Therefore, the value of the phase voltage V_{bn} is $\boxed{508.07\angle -150^\circ \text{ V}}$.

Step 6 of 6

Calculate the value of the voltage V_{cn}.

$$V_{cn} = V_{an}\angle 120^\circ$$

Substitute $508.07\angle -30^\circ$ V for V_{an}.

$$\begin{aligned} V_{cn} &= (508.07\angle -30^\circ)(1\angle 120^\circ) \\ &= 508.07\angle 90^\circ \text{ V} \end{aligned}$$

Therefore, the value of the phase voltage V_{cn} is $\boxed{508.07\angle 90^\circ \text{ V}}$.